Building Resilient Supply Chains

By Jordan Ellis, Operations Research Group — March 2026

Why resilience now

Regional distribution networks have spent the last decade optimising for cost. The result is a set of long, thin supply chains that work well in calm conditions and poorly when a single port, carrier or supplier fails. This article looks at what the most resilient operators did differently during the disruptions of the past two years.

The first difference is visibility. Operators that recovered fastest could see inventory positions across every tier of their network within hours, not weeks. They had invested in shared data models with their suppliers and treated forecast accuracy as an operational metric rather than a planning curiosity.

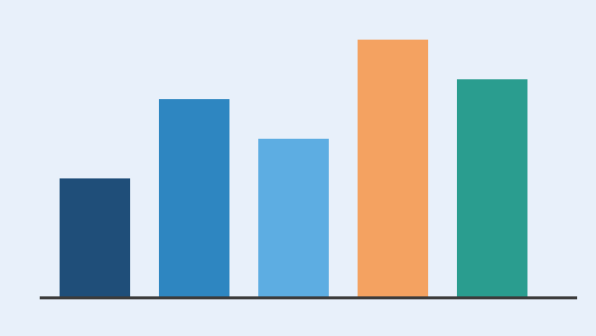

Figure 1. Recovery time by network type (weeks).

Three habits of resilient operators

The second difference is optionality. Rather than holding large buffers everywhere, resilient networks held small buffers in a few well-chosen nodes and maintained qualified alternative suppliers for their twenty most critical components. Dual sourcing cost them between two and four percent of landed cost, which most of them recovered in a single disrupted quarter.

The third difference is decision speed. Teams that had rehearsed disruption scenarios made rerouting decisions in a day. Teams that had not spent a week assembling the facts. Speed mattered more than precision: a reasonable decision taken early outperformed an optimal decision taken late in almost every case we examined.

- Shared inventory data with tier-one suppliers
- Qualified alternatives for critical parts
- Quarterly disruption rehearsals

What it costs

None of this is free. Visibility requires integration work that rarely appears on a roadmap until it is urgently needed. Optionality requires supplier development budgets. Decision speed requires authority to be delegated to people close to the problem. Each of these is an organisational change as much as a technical one.

Where to start

Our recommendation is to start with the twenty components that would stop production if they disappeared for a month. Map their supply paths to the second tier, qualify one alternative for each, and run one tabletop exercise per quarter. That programme can be delivered in under a year by a small team.

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